

Chapter 1

Preliminaries

1.1 Basic Covering Properties

We use the usual terms and notations^[1]. An ordinal is the set of all smaller ordinals, and a cardinal is an initial ordinal. We use κ, μ for cardinals; λ a for infinite cardinal; $\alpha, \beta, \gamma, \theta$ for ordinal numbers. $\omega = \{0, 1, 2, \dots\}$ is the smallest infinite cardinal. $|A|$ denotes the cardinality of a set A . The empty set ϕ is the unique set B such that $|B| = 0$. $\mathbb{R}$ denotes the real line $[0, 1]$ the closed unit interval in $\mathbb{R}$.

For any sets $Z, Y, A \subset Z$, we denote $\mathcal{P}(Z) = \{A : A \subset Z\}$; $Y^X = \{f : f \text{ is a function from } X \text{ into } Y\}$. If $f \in Y^X$, we write $f : X \rightarrow Y$.

If $f : \mathbb{Z} \rightarrow Y$, $A \subset \mathbb{Z}$ and $B \subset Y$, the $f(A) : A \rightarrow Y$ defined by $f(A(x)) = f(x)$ for each $x \in A$; $f(A) = \{f(x) : x \in A\}$. $f^{-1}B = \mathcal{F}(B) = \{x \in \mathbb{Z} : f(x) \in B\}$.

For a set A and $n \rightarrow \infty$, let $[A]^n = \{B \subset A : |B| = n\}$,

$[A]^{<w} = \cup\{[A]^n : n < w\}$.

Let $w^0 = \{\phi\}$, for $n = 1, \dots, w^n =$

$= \{f : f; n \text{ to } w\} = \{(k_0, t_1, \dots, t_{n-1}) : \forall i < n \forall i \in w\}$ where

$m_i = f(i)$. If $t < w$, let $\phi \oplus h = (\overline{t_1}), (k_0, \dots, k_{n-1}) \oplus h =$

$(\overline{k_0}, \dots, \overline{k_{n-1}}, \dots)$. Let $w^{<w} = \cup w^n$.

Let $U = \{U_\alpha : \alpha \in \Delta\}$ be any family of subsets of $\mathbb{Z}$. Without loss of generality we may assume the indexing set Δ be any set of ordinals. We call